Counseling Module

Lecture 5: Core AI

Prof. Dr. Michael Granitzer

October 15, 2025

AI-BSC-Coun-CoreAI

- •Studying at the University of Passau –
- •B.Sc. Artificial Intelligence



Slide Image

Counseling Module

Lecture 5: Core AI

October 15, 2025

Faculty of Computer Science and Mathematics

\

\

\



Slide Image

Counseling Module

Lecture 5: Core AI

October 15, 2025

Faculty of Computer Science and Mathematics

\

\

\

Counseling Module

Lecture 5: Core AI

October 15, 2025

Faculty of Computer Science and Mathematics

\

\

\

Counseling Module

Lecture 5: Core AI

October 15, 2025

Faculty of Computer Science and Mathematics

\

\

\

October 15, 2025

Faculty of Computer Science and Mathematics

\

\

\

Faculty of Computer Science and Mathematics

\

\

\

\

\

\

\

- •Study Programme from the AI perspective
- •How do modules apply to modern AI Systems
- •Core AI Modules in detail
- •Elective Modules in (more) detail
- •Questions & Answers

Outline

October 15, 2025

Faculty of Computer Science and Mathematics

\

Outline

October 15, 2025

Faculty of Computer Science and Mathematics

\

October 15, 2025

Faculty of Computer Science and Mathematics

\

Faculty of Computer Science and Mathematics

\

\

October 15, 2025

\

\

\

\

\

\

\

October 15, 2025

\

\

\

\

\

\

\

- •Study Programme from the AI perspective
- •How do modules apply to modern AI Systems
- •Core AI Modules in detail
- •Elective Modules in (more) detail
- •Questions & Answers

Outline

October 15, 2025

Faculty of Computer Science and Mathematics

\

Outline

October 15, 2025

Faculty of Computer Science and Mathematics

\

October 15, 2025

Faculty of Computer Science and Mathematics

\

Faculty of Computer Science and Mathematics

\

\



Today’s Frontier AI Method - Transformers

October 15, 2025

\



October 15, 2025

\



\







Transformer Architecture – Attention + Deep Learning

October 15, 2025

\



October 15, 2025

\



\










Slide Image


Slide Image

Theoretical Analysis of Transformers / AI Models

October 15, 2025

\

October 15, 2025

\

\





- •How to implement such a Transfomer Model?
- •Can we keep it modular and componets re-usable?
- •What data and programme structure gives us best efficiency?
- •Deploying and operating such a model?
- •How to engineer the data?
- •How to scale such systems

Realising Transformer/AI Models

October 15, 2025

\

Realising Transformer/AI Models

October 15, 2025

\

October 15, 2025

\

\



Evaluating & Applying Transformer/AI Models

October 15, 2025

\

October 15, 2025

\

\



- •How to evaluate AI Systems?
- •How to adapt it to different modalities?
- •How to build Agent systems on top of AI models?
- •Impact of AI systems and how to regulate them?
- •Wider application of AI Systems
- •…
- •How to evaluate AI Systems?
- •How to adapt it to different modalities?
- •How to build Agent systems on top of AI models?
- •Impact of AI systems and how to regulate them?
- •Wider application of AI Systems
- •…



- •Study Programme from the AI perspective
- •How do modules apply to modern AI Systems
- •Core AI Modules in detail
- •Elective Modules in (more) detail
- •Questions & Answers

Outline

October 15, 2025

Faculty of Computer Science and Mathematics

\

Outline

October 15, 2025

Faculty of Computer Science and Mathematics

\

October 15, 2025

Faculty of Computer Science and Mathematics

\

Faculty of Computer Science and Mathematics

\

\



Machine Learning

Topics

- •History of AI, basic concepts & sub-areas
- •Basics of machine learning (supervised and unsupervised learning, features)
- •Association analysis
- •Cluster analysis
- •Classification algorithms
- •Regression methods
- •Evaluation methods for learnt models

Prof. Herbold, AI Engineering

Foundations of AI Modules

October 15, 2025

Faculty of Computer Science and Mathematics

\





Topics

- •History of AI, basic concepts & sub-areas
- •Basics of machine learning (supervised and unsupervised learning, features)
- •Association analysis
- •Cluster analysis
- •Classification algorithms
- •Regression methods
- •Evaluation methods for learnt models

Prof. Herbold, AI Engineering

Foundations of AI Modules

October 15, 2025

Faculty of Computer Science and Mathematics







Foundations of AI Modules

October 15, 2025

Faculty of Computer Science and Mathematics

\





October 15, 2025

Faculty of Computer Science and Mathematics

\





\


Slide Image


Slide Image

\


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image

Probabilistic Machine Learning

Topics

- •Fundamentals of Statistical Learning and Probabilistic Machine Learning, Learning as Inference
- •Data Representation / Feature Engineering
- •Models: Naive Bayes, Probabilistic Graphical Models, Bayesian Belief Networks, Bayesian Linear Models, Mixture models
- •Model Evaluation, Model Selection, and Hyperparameter Tuning

Prof. Granitzer, Data Science

Foundations of AI Modules

October 15, 2025

\



Slide Image



Topics

- •Fundamentals of Statistical Learning and Probabilistic Machine Learning, Learning as Inference
- •Data Representation / Feature Engineering
- •Models: Naive Bayes, Probabilistic Graphical Models, Bayesian Belief Networks, Bayesian Linear Models, Mixture models
- •Model Evaluation, Model Selection, and Hyperparameter Tuning

Prof. Granitzer, Data Science

Foundations of AI Modules

October 15, 2025

\





Foundations of AI Modules

October 15, 2025

\





October 15, 2025

\





\






Slide Image


Slide Image


Slide Image

Deep Learning

Topics

- •Basics of Representation Learning
- •Perceptron LearningFeedforward Neural Networks
- •Gradient Descent and Backpropagation
- •Regularization in Deep Learning
- •Advanced Neural Network Architectures such as Convolutional and Recurrent Neural Networks
- •State-of-the-art developments in Deep Learning

Prof. Lemmerich, Applied Machine Learning

Foundations of AI Modules

October 15, 2025

\





Topics

- •Basics of Representation Learning
- •Perceptron LearningFeedforward Neural Networks
- •Gradient Descent and Backpropagation
- •Regularization in Deep Learning
- •Advanced Neural Network Architectures such as Convolutional and Recurrent Neural Networks
- •State-of-the-art developments in Deep Learning

Prof. Lemmerich, Applied Machine Learning

Foundations of AI Modules

October 15, 2025

\

Slide Image

Slide Image

Foundations of AI Modules

October 15, 2025

\

Slide Image

Slide Image

October 15, 2025

\

Slide Image

Slide Image

\


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image

Multiagent Systems

Topics

- •Distributed Optimization, Nash-Equilibrium, Congestion Games, Pricing,
- •Resource Allocation, Auctions, Mechanism Design, Learning in Games

You are able to model multiagent distributed systems with strategically interacting agents and algorithmically solve such systems by means of computing equilibrium solutions.

You learn the theory of mechanism design for distributed resource allocation problems

Prof. Harks, Mathematical Optimisation

Foundations of AI Modules

October 15, 2025

\


Slide Image



Topics

- •Distributed Optimization, Nash-Equilibrium, Congestion Games, Pricing,
- •Resource Allocation, Auctions, Mechanism Design, Learning in Games

You are able to model multiagent distributed systems with strategically interacting agents and algorithmically solve such systems by means of computing equilibrium solutions.

You learn the theory of mechanism design for distributed resource allocation problems

Prof. Harks, Mathematical Optimisation

Foundations of AI Modules

October 15, 2025

\





Foundations of AI Modules

October 15, 2025

\





October 15, 2025

\





\






Slide Image


Slide Image


Slide Image

Compulsatory Modules

October 15, 2025

\







October 15, 2025

\







\


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image

Compulsatory Modules

October 15, 2025

\



October 15, 2025

\



\










Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image

Compulsatory Modules

October 15, 2025

\



October 15, 2025

\



\










Slide Image


Slide Image


Slide Image


Slide Image

AI Project, AI Seminar & Bachelor Thesis

AI-Project

Solve a (more) complex AI problems / topics as team

Responsible: Granitzer, Harks, Herbold, Lemmerich

AI-Seminar

Individual work on deepening knowledge in a specific topic, write it up, present it and discuss with colleagues

Responsible: every professor





Bachlor Thesis

Individual work going deeper on one specific topic (either practical or theoretical). Learning to work scientifically under guidance.

Responsible: every professor

Compulsatory Modules - Your final steps

October 15, 2025

\



AI-Project

Solve a (more) complex AI problems / topics as team

Responsible: Granitzer, Harks, Herbold, Lemmerich

AI-Seminar

Individual work on deepening knowledge in a specific topic, write it up, present it and discuss with colleagues

Responsible: every professor





Bachlor Thesis

Individual work going deeper on one specific topic (either practical or theoretical). Learning to work scientifically under guidance.

Responsible: every professor

Compulsatory Modules - Your final steps

October 15, 2025

\



AI-Seminar

Individual work on deepening knowledge in a specific topic, write it up, present it and discuss with colleagues

Responsible: every professor





Bachlor Thesis

Individual work going deeper on one specific topic (either practical or theoretical). Learning to work scientifically under guidance.

Responsible: every professor

Compulsatory Modules - Your final steps

October 15, 2025

\







Bachlor Thesis

Individual work going deeper on one specific topic (either practical or theoretical). Learning to work scientifically under guidance.

Responsible: every professor

Compulsatory Modules - Your final steps

October 15, 2025

\





Bachlor Thesis

Individual work going deeper on one specific topic (either practical or theoretical). Learning to work scientifically under guidance.

Responsible: every professor

Compulsatory Modules - Your final steps

October 15, 2025

\



Bachlor Thesis

Individual work going deeper on one specific topic (either practical or theoretical). Learning to work scientifically under guidance.

Responsible: every professor

Compulsatory Modules - Your final steps

October 15, 2025

\



Compulsatory Modules - Your final steps

October 15, 2025

\



October 15, 2025

\



\





- •Study Programme from the AI perspective
- •How do modules apply to modern AI Systems
- •Core AI Modules in detail
- •Elective Modules in (more) detail
- •Questions & Answers

Outline

October 15, 2025

Faculty of Computer Science and Mathematics

\

Outline

October 15, 2025

Faculty of Computer Science and Mathematics

\

October 15, 2025

Faculty of Computer Science and Mathematics

\

Faculty of Computer Science and Mathematics

\

\



Elective Modules

October 15, 2025

\

October 15, 2025

\

\

You choose through

- •16 ECTS from the chosen compulsory elective group
- •45 ECTS from three elective groups
- ▪ –Each elective group must comprise at least 10 ECTS;
- –At least one elective group must come from Computer Science or Mathematics.
- –German as a Foreign Language with at least 20 ECTS

(recommended, if you want to establish yourself in Germany)

- •Details in the Module Catalog:



Compulsory Electivse – Advaced AI Applied

October 15, 2025

\

October 15, 2025

\

\














Slide Image


Slide Image


Slide Image


Slide Image

Compulsory Electives – Advanced AI Theory

October 15, 2025

Faculty of Computer Science and Mathematics

\



October 15, 2025

Faculty of Computer Science and Mathematics

\



Faculty of Computer Science and Mathematics

\



\


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image

Elective Modules – Computer Science

October 15, 2025

\

October 15, 2025

\

\


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image

Elective Modules – Computer Science

October 15, 2025

\



October 15, 2025

\



\










Slide Image


Slide Image


Slide Image


Slide Image


Slide Image

Elective Modules – Mathematics

October 15, 2025

\



October 15, 2025

\



\










Slide Image


Slide Image


Slide Image


Slide Image


Slide Image

Elective Modules – Mathematics

\



\


















Slide Image


Slide Image


Slide Image


Slide Image


Slide Image


Slide Image

- •Study Programme from the AI perspective
- •How do modules apply to modern AI Systems
- •Core AI Modules in detail
- •Elective Modules in (more) detail
- •Questions & Answers

Outline

October 15, 2025

Faculty of Computer Science and Mathematics

\

Outline

October 15, 2025

Faculty of Computer Science and Mathematics

\

October 15, 2025

Faculty of Computer Science and Mathematics

\

Faculty of Computer Science and Mathematics

\

\

Join here:

- •https://www.menti.com/al1m3pzeozro

Questions and Answers

October 15, 2025

\



Questions and Answers

October 15, 2025

\



October 15, 2025

\


Slide Image

\


Slide Image


Slide Image

October 15, 2025

Faculty of Computer Science and Mathematics

\

Any Questions?

Faculty of Computer Science and Mathematics

\

Any Questions?

\

Any Questions?

Any Questions?

